

Rate Constants from Monte Carlo Quasi-Classical Trajectory Calculations. A Procedure for Importance Sampling^{1a}

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The use of importance sampling as a variance reduction technique is applied to the *direct* Monte Carlo quasi-classical trajectory (MCQT) calculation of bimolecular gas phase rate constants. A general importance sampling function for the collision energy having a single (integer) parameter which characterizes the distribution of collision energies to be sampled is proposed and used in calculations of rate constants for three chemical reactions [$F + H_2$, $Cl + H_2$, and $O(^3P) + H_2$] at several temperatures. Results are compared with regard to accuracy and efficiency with rate constants obtained directly or indirectly using other MCQT approaches and the same potential energy surfaces. The importance sampling calculations are found to be more efficient over the entire range of reactivity studied, and provide statistically valid estimates of the errors in the computed rate constants.

I. Introduction

The most commonly used procedure for obtaining rate constants from Monte Carlo quasi-classical trajectory (MCQT) calculations may be characterized as *indirect*.² First, the reaction cross section for each significantly populated internal state of the reactants is computed at several collision energies. These results are then fitted with some chosen functional forms to obtain excitation functions for each different reactant internal state considered. Finally, these excitation functions are averaged over the appropriate distributions of collision energies and reactant internal states to obtain the desired rate constant. Points *in favor* of such an approach are as follows: (1) one set of excitation functions, once obtained, contains enough information for calculating the rate constant for the reaction at an arbitrarily large number of temperatures, and (2) sampling of collision energies below the threshold for reaction, once it is established, is unnecessary, thereby increasing the efficiency of calculating rate constants for reactions with high energy thresholds. Several disadvantages of this indirect approach are (1) that accurate thresholds are difficult to determine in MCQT calculations; (2) to calculate enough information for determining accurate excitation functions is often to calculate much information to which the rate constants are insensitive, thereby reducing the efficiency of the calculation; and (3) it is difficult to determine a reasonable estimate of the error in the computed rate constants when the directly computed quantities (cross sections) are so far removed from the final result.

An alternative approach is to compute the desired rate constant *directly* by using the MCQT method to evaluate the multidimension integral expression relating the rate constant to the reaction probability, i.e., by including the collision energy and reactant internal states in the set of randomly selected initial variables.³ Here the collision energy is chosen from the same distribution function used in the averaging of excitation functions in the indirect method. While such a direct method does allow a statistically valid estimate of the error in the computed rate constant, it is feasible only for reactions with low thresholds at high temperatures, or other "fast reactions". This is because in other cases the total number of trajectories required to obtain a sufficiently large number of

them with the collision energy greater than the threshold value becomes prohibitively large.

In the present work, we examine the applicability of the variance reduction technique of importance sampling to the *direct* MCQT calculation of bimolecular gas phase rate constants and *their estimated errors*. It will be seen that this approach is more efficient than either alternative discussed above, and, when used appropriately, produces accurate rate constants with reliable error estimates. A general importance sampling function for the collision energy having a single (integer) parameter is proposed and tested. Guidelines are provided for selecting the optimum value of this parameter for a given chemical system and temperature. Diagnostics for determining the quality of the importance sampling used in a calculation are discussed. A procedure for obtaining other quantities (e.g., the excitation function) from an importance sampling calculation of the rate constant is illustrated.

The paper is organized as follows. The general considerations of importance sampling are discussed in section II.A and are applied to the MCQT calculation of bimolecular gas phase rate constants in section II.B. The importance sampling approach is then tested by calculating rate constants for three well-characterized chemical reactions at several temperatures. The results of these calculations are presented in section III and are compared with regard to accuracy and efficiency with other directly or indirectly computed rate constants using the MCQT method and the same potential energy surfaces. A summary and some suggested refinements of the methods employed in the present work are presented in section IV.

II. Importance Sampling

Importance sampling is a well-known technique in the Monte Carlo evaluation of a (multidimensional) integral which reduces the sample variance and hence reduces the estimated error.⁴ In section II.A we will review the importance sampling technique as it applies to MCQT calculations in general.⁵ Section II.B is devoted exclusively to the use of the importance sampling technique in the selection of the impact parameter and relative energy in the direct MCQT calculation of bimolecular rate constants.

A. *Importance Sampling in MCQT Calculations.* Consider the reaction probability, $P_r(x,q)$, which is a

function of some specific variable x and some other set of variables $\mathbf{q}$. Here the function P_r may exhibit one of only two possible values, either 0 or 1, for a given set of reactant variables, i.e., it is a Boolean function. Next consider some integral, say I , of the function P_r over various distribution functions of the initial variables, i.e.

$$I \equiv C \int_{x_{\min}}^{x_{\max}} \int_{q_{1,\min}}^{q_{1,\max}} \dots \int_{q_{h,\min}}^{q_{h,\max}} P_r(x, \mathbf{q}) g(x) \times h_1(q_1) \dots h_h(q_h) dx d\mathbf{q} \quad (1)$$

Here $g(x)$ is the weighting function for the variable x , $h_i(q_i)$ is that for each of the variables q_i , and C is a constant. The crude Monte Carlo approach to the evaluation of eq 1 would be to transform that equation to the form

$$I = C \int_{u_x=0}^1 \int_{u_{q_1}=0}^1 \dots \int_{u_{q_h}=0}^1 P_r[x(u_x), \mathbf{q}(\mathbf{u}_q)] du_x d\mathbf{u}_q \quad (2)$$

by defining

$$du_x = g(x) dx \equiv dG \quad (3a)$$

and

$$du_{q_i} = h_i(q_i) dq_i \equiv dH_i \quad (3b)$$

which gives

$$u_x = \int_{x_{\min}}^x g(x) dx \equiv \int_{G(x_{\min})}^{G(x)} dG \quad (4a)$$

and

$$u_{q_i} = \int_{q_{i,\min}}^{q_i} h_i(q_i) dq_i \equiv \int_{H_i(q_{i,\min})}^{H_i(q_i)} dH_i \quad (4b)$$

where we have assumed the normalization conditions on the functions $g(x)$ and $h_i(q_i)$ which give

$$u_x(x_{\max}) = \int_{x_{\min}}^{x_{\max}} g(x) dx = 1 \quad (5a)$$

and

$$u_{q_i}(q_{i,\max}) = \int_{q_{i,\min}}^{q_{i,\max}} h_i(q_i) dq_i = 1 \quad (5b)$$

Note that any multiplicative constants required for these normalizations are taken up by the constant C . Then a crude Monte Carlo estimator of the integral I is obtained by averaging $P_r(u_x, \mathbf{u}_q)$ over N randomly selected points in the $(k+1)$ -dimensional unit hypercube

$$I = C \langle P_r \rangle \quad (6a)$$

$$\approx \frac{C}{N} \sum_{j=1}^N P_{rj} \quad (6b)$$

$$= \frac{C}{N} \sum_{j=1}^{N_r} P_{rj} = C \frac{N_r}{N} \equiv \tilde{I} \quad (6c)$$

where P_{rj} represents the value of P_r at the j th randomly selected point in the unit hypercube and the sum in eq 6c is over only the N_r reactive trajectories (P_{rj} is zero for nonreactive trajectories). The corresponding Monte Carlo estimator for the variance of this estimator for I is

$$\text{var}(\tilde{I}) = \frac{C^2}{N} \{ \langle P_r^2 \rangle - \langle P_r \rangle^2 \} \quad (7a)$$

$$\approx \frac{C^2}{N} \left\{ \frac{1}{N} \sum_{j=1}^N P_{rj}^2 - \left(\frac{1}{N} \sum_{j=1}^N P_{rj} \right)^2 \right\} \quad (7b)$$

$$= \frac{C^2}{N} \left\{ \frac{N_r}{N} - \frac{N_r^2}{N^2} \right\} = C^2 \frac{N_r^2}{N^2} \left(\frac{N - N_r}{N_r N} \right) \quad (7c)$$

so that the estimated error in the value $\tilde{I}$ is (at the 68% confidence level)

$$\Delta \tilde{I} = [\text{var}(\tilde{I})]^{1/2} \quad (8b)$$

$$\approx C \frac{N_r}{N} \left(\frac{N - N_r}{N_r N} \right)^{1/2} \quad (8b)$$

As is seen in eq 7a, the $\text{var}(\tilde{I})$ is directly proportional to the $\text{var}(P_r)$, i.e., to the variance of the integrand in eq 2. If this integrand exhibits appreciable structure, then its variance will be rather large. The importance sampling technique attempts to transform this integrand to some new function which is (ideally) nearly uniform over the unit hypercube.^{4,5} The new integrand would thus exhibit a greatly reduced variance and lead to a correspondingly reduced value for the estimated error in $\tilde{I}$.

The transformed integrand corresponding to importance sampling on the variable x is obtained by multiplying the integrand in eq 1 by

$$1 = \frac{P_r^0(x)}{P_r^0(x)} \quad (9)$$

where $P_r^0(x)$ is some function which is believed to exhibit the same structure as the function

$$P_r(x) \equiv \int_{q_{1,\min}}^{q_{1,\max}} \int_{q_{2,\min}}^{q_{2,\max}} \dots \int_{q_{h,\min}}^{q_{h,\max}} P_r(x, \mathbf{q}) d\mathbf{H}(\mathbf{q}) \quad (10)$$

i.e., the same structure as the reaction probability $P_r(x, \mathbf{q})$ averaged over all the q_i .⁵ The resulting expression is then regrouped to yield

$$I = C \int_{x_{\min}}^{x_{\max}} \int_{q_{1,\min}}^{q_{1,\max}} \dots \int_{q_{h,\min}}^{q_{h,\max}} \frac{P_r(x, \mathbf{q})}{P_r^0(x)} P_r^0(x) g(x) d\mathbf{H}(\mathbf{q}) \quad (11a)$$

$$= C \int_{u_x=0}^1 \int_{u_{q_1}=0}^1 \dots \int_{u_{q_h}=0}^1 \mathcal{P}_r(u_x, \mathbf{u}_q) du_x d\mathbf{u}_q \quad (11b)$$

where

$$\mathcal{P}_r(u_x, \mathbf{u}_q) \equiv \frac{P_r[x(u_x), \mathbf{q}(u_q)]}{P_r^0[x(u_x)]} \quad (11c)$$

$$du_x = P_r^0(x) g(x) dx \quad (11d)$$

$\mathbf{u}_q$ is as above (see eq 4b), and $P_r^0(x)$, the importance sampling function, is normalized such that

$$u(x_{\max}) = \int_{x_{\min}}^{x_{\max}} P_r^0(x) g(x) dx = 1 \quad (11e)$$

The more closely $P_r^0(x)$ resembles $P_r(x)$, the lower the variance of the transformed integrand $\mathcal{P}_r(u_x, \mathbf{u}_q)$. It should be pointed out, however, that even if $P_r^0(x)$ were chosen to be exactly proportional to $P_r(x)$ as defined in eq 10, the variance of $\mathcal{P}_r$ would not in general be zero. This is because the value of $P_r(x)$ at any value of x is an average of ones and zeros which depend on the other integration variables, u_{q_i} . The choice of $P_r^0(x) \propto P_r(x)$ is, however, the best importance sampling function of the single variable x . A lower variance of $\mathcal{P}_r$ can be obtained only by importance sampling on other variables independently, which is an obvious extension of the foregoing discussion, or by using a correlated (nonseparable) importance sampling function (e.g., specifying a different value of the maximum impact parameter, $b_{\max}(v)$, for each reactant vibrational state, v).

The Monte Carlo estimator for I in eq 11 is

$$I \approx \tilde{I} = \frac{C}{N} \sum_{j=1}^N \mathcal{P}_{rj} \quad (12a)$$

$$= \frac{C}{N} \frac{\sum_{j=1}^N \mathcal{P}_{rj}}{\sum_{j=1}^N \mathcal{P}_{rj}^0} \quad (12b)$$

$$\equiv \frac{C}{N} \sum_{j=1}^N \mathcal{P}_{rj} w_j \quad (12c)$$

$$= \frac{C}{N} \sum_{j=1}^{N_r} w_j \quad (12d)$$

where we have identified the *statistical weight*, w_j , of each trajectory to be $1/\mathcal{P}_{rj}^0$.

The estimated error in $\tilde{I}$ in a MCQT calculation with importance sampling is obtained from⁵

$$\text{var}(\tilde{I}) = \frac{C^2}{N} \text{var}(\mathcal{P}_r) \quad (13a)$$

$$= \frac{C^2}{N} \{ \langle \mathcal{P}_r^2 \rangle - \langle \mathcal{P}_r \rangle^2 \} \quad (13b)$$

$$\approx \frac{C^2}{N} \left\{ \frac{1}{N} \sum_{j=1}^{N_r} w_j^2 - \left(\frac{1}{N} \sum_{j=1}^{N_r} w_j \right)^2 \right\} \quad (13c)$$

$$= \frac{C^2}{N^2} \left\{ \sum_{j=1}^{N_r} w_j^2 - \frac{1}{N} \left(\sum_{j=1}^{N_r} w_j \right)^2 \right\} \quad (13d)$$

and is

$$\Delta \tilde{I} \approx \frac{C}{N} \left\{ \sum_{j=1}^{N_r} w_j^2 - \frac{1}{N} \left(\sum_{j=1}^{N_r} w_j \right)^2 \right\}^{1/2} \quad (14)$$

B. Application to the MCQT Calculation of Bimolecular Rate Constants. We will deal explicitly in this section only with rate constants for reactions of an atom with specific vibrational states of a diatomic molecule. The generalizations of this presentation to fully averaged (over reactant vibrational states) rate constants and to molecule-molecule reactions should become obvious. We will designate the specific rate constant for an atom-diatomic molecule reaction as $k_r^v(T)$, where v labels the vibrational state of the molecule and T is the temperature. The relation between $k_r^v(T)$ and the reaction probability is⁵

$$k_r^v(T) = \left(\frac{8kT}{\pi\mu} \right)^{1/2} \pi b_{\max}^2 [(2\pi)^3 b_{\max}^2 (kT)^2]^{-1} \sum_{j=0}^{j_{\max}} \frac{g_j (2j+1) e^{-\epsilon_{vj}/kT}}{Q_{rv}} \times \int_{E_{\text{rel}}=0}^{\infty} \int_{b=0}^{b_{\max}} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \int_{\eta=0}^{2\pi} \int_{\xi=0}^{2\pi} \mathcal{P}_r(j, E_{\text{rel}}, b, \theta, \phi, \eta, \xi; v) \times E_{\text{rel}} e^{-E_{\text{rel}}/kT} b \sin \theta dE_{\text{rel}} db d\theta d\phi d\eta d\xi \quad (15)$$

where k is the Boltzmann constant, μ is the reduced mass for the relative motion of the reactants, $b_{\max}$ is the maximum impact parameter (beyond which the reaction probability is zero), $j_{\max}$ is the maximum rotational state of the reactant molecule considered, g_j is the nuclear degeneracy (if applicable) of the j th rotational state of the reactant molecule, Q_{rv} is the rotational partition function (with the sum truncated at $j_{\max}$) for the v th vibrational state of the reactant molecule, E_{rel} is the initial relative collision energy of reactants, and b , θ , ϕ , η , and ξ are the impact parameter, azimuthal and polar orientation angles for the diatomic molecular axis, angular momentum orientation angle, and vibrational phase angle for the diatomic

molecule, respectively. The (Boolean) reaction probability function $\mathcal{P}_r(j, E_{\text{rel}}, b, \theta, \phi, \eta, \xi; v)$ is a function of the first seven variables with a fixed value of the reactant molecule vibrational quantum number, v .

The primary variable of interest with respect to importance sampling is the collision energy, E_{rel} . This is best seen by carrying out an implicit integration (or sum) over all variables other than E_{rel} in eq 15. The result is

$$k_r^v(T) = \left(\frac{8kT}{\pi\mu} \right)^{1/2} \pi b_{\max}^2 \frac{1}{(kT)^2} \int_0^{\infty} \mathcal{P}_r(E_{\text{rel}}; v) E_{\text{rel}} e^{-E_{\text{rel}}/kT} dE_{\text{rel}} \quad (16)$$

Consider a reaction with a moderately large threshold energy $E_{\text{rel}}^{\text{th}} \gg kT$ for which $\mathcal{P}_r(E_{\text{rel}}; v) = 0$ for all $E_{\text{rel}} < E_{\text{rel}}^{\text{th}}$. For such a reaction the integrand in eq 16 will be appreciable only for collision energies above $E_{\text{rel}}^{\text{th}}$, but not very far above $E_{\text{rel}}^{\text{th}}$ because of the $e^{-E_{\text{rel}}/kT}$ factor. Without importance sampling, the collision energy would be selected from the distribution function $x e^{-x}$, where $x \equiv E_{\text{rel}}/kT$, which peaks at $x = 1$ (or, $E_{\text{rel}} = kT$). Consequently, we see that not only would there be undesirable structure in the integrand, but that very few (if any) collision energies in excess of $E_{\text{rel}}^{\text{th}}$ would be sampled in a crude Monte Carlo approach to the calculation of this rate constant.

Equation 16 may be regarded as having been transformed for an importance sampling calculation in which the importance sampling function is $\mathcal{P}_r^0(E_{\text{rel}}; v) = 1$, i.e., that the expected reaction probability function is independent of E_{rel} . We know, however, that this is not the case for reactions with a threshold, as was discussed above. As one does not usually know the value of $E_{\text{rel}}^{\text{th}}$ in advance, care must be taken in selecting any other function for $\mathcal{P}_r^0(E_{\text{rel}}; v)$. In the present work we have used the function

$$\mathcal{P}_r^0(E_{\text{rel}}; v) = \frac{1}{m!} \left(\frac{E_{\text{rel}}}{kT} \right)^{m-1} \quad (17)$$

where m is some integer ≥ 1 , as a universal importance sampling function for the collision energy. The optimum choice of m will be discussed below, and will be different for different chemical systems and for the same chemical system at different temperatures.

Using the importance sampling function given by eq 17, the rate constant expression in eq 16 may be rewritten as

$$k_r^v(T) = \left(\frac{8kT}{\pi\mu} \right)^{1/2} \pi b_{\max}^2 \int_0^{\infty} \frac{\mathcal{P}_r(x; v)}{\mathcal{P}_r^0(x; v)} \frac{1}{m!} x^m e^{-x} dx \quad (18a)$$

$$= \left(\frac{8kT}{\pi\mu} \right)^{1/2} \pi b_{\max}^2 \int_0^1 \mathcal{P}_r(u; v) du \quad (18b)$$

where

$$x \equiv E_{\text{rel}}/kT \quad (18c)$$

$$du = \frac{1}{m!} x^m e^{-x} dx \equiv f_m(x) dx \quad (18d)$$

and

$$u = \int_0^u du = \int_0^x f_m(x) dx = \frac{1}{m!} \int_0^x x^m e^{-x} dx \quad (18e)$$

$$= 1 - e^{-x} \sum_{l=0}^m \frac{x^l}{l!} \quad (18f)$$

In eq 18d, $f_m(x)$ is the *biased* distribution function from

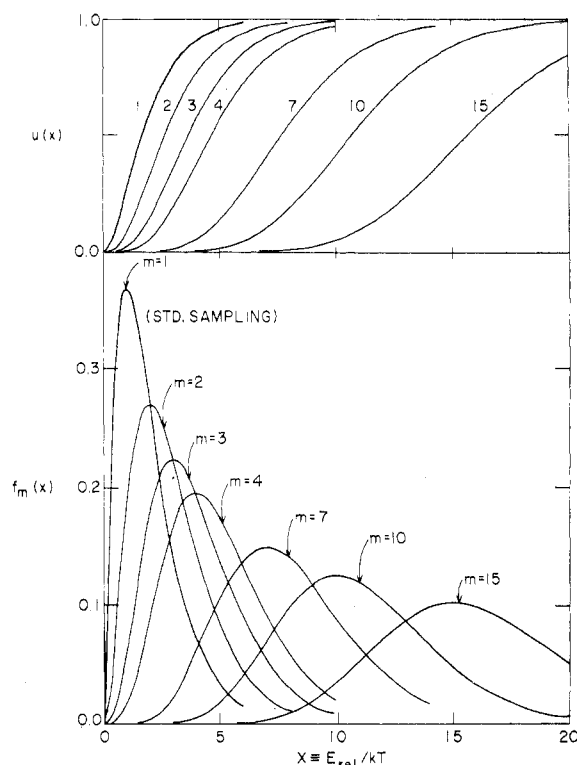


Figure 1. Plot of the distribution function for collision energies resulting from the use of the importance sampling function in eq 17 for several values of the (integer) parameter m (lower panel), and the corresponding values of the uniform variable u (upper panel). The case of $m = 1$ corresponds to standard, or crude Monte Carlo, sampling of the collision energy. The effect of choosing $m > 1$ is to bias this distribution function to larger values of the collision energy. Note that the inflection point in each $u(x)$ curve and the maximum of the corresponding $f_m(x)$ curve occur at $x = m$, where $x \equiv E_{\text{rel}}/kT$.

which the collision energy is to be sampled. This bias is removed from the computed rate constant by the consideration of the corresponding weighting function

$$w(x;v) = \frac{m!}{x^{m-1}} \quad (19)$$

The distribution function $f_m(x)$ is shown in the lower panel of Figure 1 for several values of m . Shown in the upper panel of Figure 1 are the corresponding values of $u(x)$. It is obvious from this figure that importance sampling with a given value of m in eq 18 causes the reduced collision energy, x , to be selected most frequently in the region around $x = m$. In fact, the function $f_m(x)$ peaks at $x = m$ and for $m \neq 1$ has inflection points at $m \pm m^{1/2}$. Note that $u(x)$ has an inflection point at $x = m$. This property proves quite useful in inverting the function $u(x)$ to obtain $x(u)$, or $E_{\text{rel}}(u) = kTx(u)$, as is shown in the Appendix.

Turning now to the optimum choice of m for a given situation, we need to estimate the classical threshold for the reaction. In the absence of any dynamical information we may wish to use knowledge of the potential energy surface (i.e., height and location of barriers) and qualitative notions about the availability of the vibrational energy (in the v th vibrational state) for surmounting the potential energy barrier.⁶ After $E_{\text{rel}}^{\text{th}}$ has been estimated by such considerations, a reasonable value of m may be obtained as that value for which the left-hand inflection point in $f_m(x)$ occurs at or just below $x_{\text{th}} = E_{\text{rel}}^{\text{th}}/kT$. As this inflection point occurs at $m - m^{1/2}$, this gives

$$\tilde{m} = \frac{1}{4} \{ 1 + (1 + 4x_{\text{th}})^{1/2} \}^2 \quad (20)$$

where $\tilde{m}$ is a continuous variable. The desired value of m is obtained either as the integer part of $\tilde{m}$ or by rounding

m to the nearest integer. Of these alternatives it is better to choose the one which gives $(m - m^{1/2})kT$ closer to $E_{\text{rel}}^{\text{th}}$ (slightly below rather than slightly above $E_{\text{rel}}^{\text{th}}$ is better in most cases). Note that $m = 1$ corresponds to crude Monte Carlo sampling (standard sampling) and will be optimum only for very low thresholds or very high temperatures.

The Monte Carlo estimator for the rate constant integral expression in eq 18 is

$$k_r^v(T) \approx \left(\frac{8kT}{\pi\mu} \right)^{1/2} \pi b_{\text{max}}^2 \frac{1}{N} \sum_{j=1}^{N_r} w_j \quad (21a)$$

with

$$w_j = \frac{m!}{x^{m-1}(u_j)} \quad (21b)$$

if importance sampling is carried out only for the collision energy. If other variables are importance sampled (in the present work, the impact parameter is also importance sampled⁵), the total statistical weight is the product of the statistical weights corresponding to each (independently) importance sampled variable. The estimated error in $k_r^v(T)$ is given by (see eq 7b)

$$\Delta k_r^v(T) \approx \left(\frac{8kT}{\pi\mu} \right)^{1/2} \pi b_{\text{max}}^2 \frac{1}{N} \left\{ \sum_{j=1}^{N_r} w_j^2 - \frac{1}{N} \left(\sum_{j=1}^{N_r} w_j \right)^2 \right\}^{1/2} \quad (22)$$

Upon completing an MCQT calculation of $k_r^v(T)$ and $\Delta k_r^v(T)$ with importance sampling, two questions which naturally arise are the following: (1) How optimum was the choice of m , and/or how appropriate was the importance sampling function used? (2) Does the direct calculation of $k_r^v(T)$ produce any information about other reaction attributes, e.g., the excitation function? The answer to the first question can be obtained by turning to the concepts of information theory. In particular, the deviation of $P_r(u_{E_{\text{rel}}})$ binned in equal increments of $u_{E_{\text{rel}}}$ (and averaged over all other variables of integration) from the constant function $\langle P_r \rangle$ may be quantified in terms of the value of

$$\text{DS}_n = \left(\frac{R}{\sum_{k=1}^n A_k} \right) \sum_{k=1}^n A_k \ln A_k - R \ln \left(\frac{1}{n} \sum_{k=1}^n A_k \right) \geq 0 \quad (23)$$

where $A_k \equiv \sum_{i=1}^{N_{rk}} w_i$, N_{rk} is the number of reactive trajectories in the k th bin, and n refers to the number of bins in the histogrammic representation of P_r .^{5,7} The lower the value of DS_n , the more closely $P_r(u_{E_{\text{rel}}})$ resembles the ideal constant function. The optimum value of m would be expected to yield the smallest value of DS_n . If this smallest value is still quite large, the fault probably lies in the importance sampling function given by eq 17. The answer to the second question is that indeed other reaction attributes can be determined from such a direct calculation of the rate constant. For example, a histogrammic representation of the excitation function which reveals its global behavior is readily obtained. Using the Method A-Analytical approach of Faist et al.⁵ with the partition of the collision energy axis determined by equal increments of $u_{E_{\text{rel}}}$, the value of the reaction cross section for the v th vibrational state of the reactant molecule in the k th collision energy bin is given by

$$\sigma_{rk}^v \approx \frac{\pi b_{\text{max}}^2}{N \Delta G_{kj=1}} \sum_{j=1}^{N_{rk}} w_j \quad (24a)$$

where

$$\Delta G_k = -e^{-x}(x+1)^{\frac{x(u=k/n)}{x(u=k-1/n)}} \quad (24b)$$

The estimated error in σ_{rk}^v is

$$\Delta \sigma_{rk}^v \approx \frac{\pi b_{\max}^2}{N \Delta G_k} \left\{ \sum_{j=1}^{N_{rk}} w_j^2 - \frac{1}{N} \left(\sum_{j=1}^{N_{rk}} w_j \right)^2 \right\}^{1/2} \quad (25)$$

where N_{rk} is the number of reactive trajectories with initial collision energy in the k th bin.

Still another question may arise. (3) How much more efficient was the importance sampling calculation than a crude Monte Carlo calculation would have been, or how trajectories would have been required using the crude Monte Carlo approach to obtain the same relative error as that using importance sampling? In a crude Monte Carlo calculation the relative error is given by

$$\frac{\Delta k_r^v(T)}{k_r^v(T)} = \frac{\Delta \langle P_r \rangle}{\langle P_r \rangle} = \left[\frac{(N - N_r)}{N_r N} \right]^{1/2} = \left[\frac{(1 - \langle P_r \rangle)}{N \langle P_r \rangle} \right]^{1/2} \quad (26)$$

If this equation is solved for N in terms of $\langle P_r \rangle$ and $\Delta \langle P_r \rangle$, and the values of $\langle P_r \rangle$ and $\Delta \langle P_r \rangle$ from the importance sampling calculation are substituted into that expression, an estimate of the number of trajectories required for the same accuracy

$$\tilde{N} = \frac{\langle P_r \rangle (1 - \langle P_r \rangle)}{(\Delta \langle P_r \rangle)^2} \quad (27)$$

is obtained.

III. Results and Discussion

To test both the accuracy and efficiency of the importance sampling technique as applied to the direct MCQT calculation of rate constants for atom-diatom molecule reactions, we have recomputed rate constants for two systems for which reliable rate constants have been calculated *indirectly* using the MCQT method. In both cases the previous workers first calculated excitation functions for each significantly populated rotational state (the vibrational states were fixed), then the MCQT results were fitted with some choice of analytical functions, which were integrated and Boltzmann averaged to obtain $k_r^v(T)$. Persky⁸ applied this procedure to the reaction $\text{Cl} + \text{H}_2(v=0) \rightarrow \text{HCl} + \text{H}$, and Johnson and Winter⁹ applied it to the reaction $\text{O}(^3\text{P}) + \text{H}_2(v=0) \rightarrow \text{OH} + \text{H}$. Generalized LEPS potential energy surfaces were used in both studies, and in each case we have employed the same potential surface used in the original work. The barriers to reaction on these potential surfaces are 7.7 and 12.5 kcal mol⁻¹ for the $\text{Cl} + \text{H}_2$ and $\text{O}(^3\text{P}) + \text{H}_2$ systems, respectively. We have computed direct MCQT rate constants for these reactions using the importance sampling technique at those temperatures for which rate constant values were reported in the original studies.^{8,9} We have also calculated rate constants for the reaction $\text{F} + \text{H}_2(v=0) \rightarrow \text{HF} + \text{H}$ at several temperatures using the importance sampling technique. For the potential energy surface for the $\text{F} + \text{H}_2$ system we have used Muckerman's¹⁰ Surface 5, on which the barrier to reaction is 1.1 kcal mol⁻¹. Thus the three systems chosen for testing the application of the importance sampling technique to the direct MCQT calculation of rate constants span a very large range of reactivity.

Prior to any calculations to be used in a comparison with those of the original studies of the chosen systems, we carried out preliminary studies to develop diagnostics for

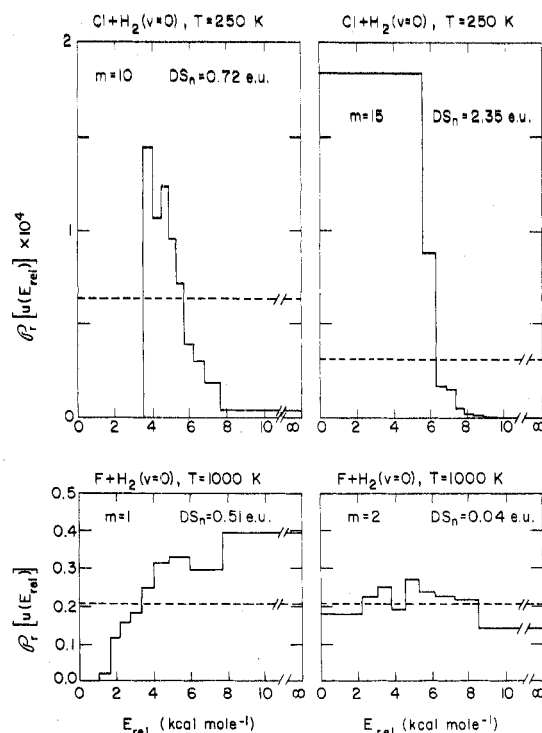


Figure 2. Plots of the histogrammic representations of $\mathcal{P}_r[u(E_{\text{rel}})]$ (solid lines) for ten equal increments of u from importance sampling calculations of the rate constants for the $\text{F} + \text{H}_2(v=0)$ system at $T = 1000$ K (lower two panels) and the $\text{Cl} + \text{H}_2(v=0)$ system at $T = 250$ K (upper two panels). Results for two different choices of the importance sampling parameter, m , are shown for each system. The dashed lines indicate the ideal (constant) result expected from a perfect choice of importance sampling function in the absence of statistical noise, and correspond to $\mathcal{P}_r[u(E_{\text{rel}})] = \langle P_r \rangle$. The DS_n values are also indicated in each panel.

good and poor choices of the importance sampling parameter, m . The results of these preliminary considerations are exemplified in Figure 2. Each panel in the figure shows a ten-bin histogrammic representation of $\mathcal{P}_r[u(E_{\text{rel}})]$ for equal increments of u , and the corresponding value of $\langle P_r \rangle$.⁵ The lower two panels refer to importance sampling calculations of the rate constant for the $\text{F} + \text{H}_2(v=0)$ reaction at $T = 1000$ K using two different values of m . The upper two panels refer to similar calculations for the $\text{Cl} + \text{H}_2(v=0)$ reaction at $T = 250$ K, again using two (this time very different) values of m . For the $\text{F} + \text{H}_2(v=0)$ system the values of $\langle P_r \rangle$ corresponding to the two choices of m are quite similar, but $\mathcal{P}_r[u(E_{\text{rel}})]$ for the case of $m = 2$ is more nearly constant than it is for $m = 1$. This feature is reflected by the DS_n values, which show importance sampling with $m = 2$ to be nearly perfect for this system at this temperature. The situation is somewhat different in the case of the $\text{Cl} + \text{H}_2(v=0)$ system, in which it is seen that the two values for $\langle P_r \rangle$ corresponding to the two different choices of m are markedly dissimilar. For neither choice of m does $\mathcal{P}_r[u(E_{\text{rel}})]$ appear flat (constant), but the DS_n values strongly imply that the choice of $m = 10$ is better than $m = 15$. The sharp structure in $\mathcal{P}_r[u(E_{\text{rel}})]$ for $m = 10$ is indicative of a general feature of low temperature, large m importance sampling using eq 18: $f_m(x)$ is generally broader than the integrand in eq 16 so that the best value of m corresponds to aligning the peak of $f_m(x)$ with the peak of the integrand as nearly as possible. A diagnostic of the proper alignment is that the peak in $\mathcal{P}_r[u(E_{\text{rel}})]$ should occur in one of the narrowest energy bins in a plot such as those in Figure 2. This is indeed the case for $m = 10$ in the $\text{Cl} + \text{H}_2(v=0)$ system. The distinctly non-zero value of DS_n for this optimum choice of m reflects

TABLE I: Parameters, Results, and Diagnostics of Direct Importance Sampling Calculations of Rate Constants

T, K	$f(T)$	m	$k_r(T)^a$	$k_{r, \text{other}}(T)^{a, b}$	$\bar{N}$	DS_n, eu
$F + H_2(v=0) \rightarrow HF + H$						
200	0.487	4	$(2.0_1 \pm 0.1_8) \times 10^{-12}$		5.47×10^3	0.41
250	0.477	3	$(4.0_3 \pm 0.3_1) \times 10^{-12}$		4.18×10^3	
300	0.466	3	$(6.2_5 \pm 0.4_3) \times 10^{-12}$		3.44×10^3	0.38
500	0.432	2	$(1.4_5 \pm 0.0_9) \times 10^{-11}$		2.24×10^3	
1000	0.391	2	$(3.4_8 \pm 0.1_6) \times 10^{-11}$		1.78×10^3	0.04
$Cl + H_2(v=0) \rightarrow HCl + H$						
250	0.498	10	$(5.3_5 \pm 1.5_2) \times 10^{-15}$	5.74×10^{-15c}	1.96×10^5	0.72
300	0.496	9	$(1.8_3 \pm 0.3_3) \times 10^{-14}$	2.29×10^{-14c}	1.55×10^5	1.13
400	0.490	7	$(1.1_0 \pm 0.1_7) \times 10^{-13}$	1.27×10^{-13c}	3.93×10^4	
500	0.481	6	$(3.7_9 \pm 0.4_4) \times 10^{-13}$	3.99×10^{-13c}	2.28×10^4	0.75
600	0.471	5	$(7.4_4 \pm 0.8_2) \times 10^{-13}$	8.53×10^{-13c}	1.37×10^4	0.51
1000	0.438	4	$(4.8_4 \pm 0.4_5) \times 10^{-12}$		3.40×10^3	0.20
$O(^3P) + H_2(v=0) \rightarrow OH + H$						
300	0.811	18	$(5.8_4 \pm 4.4_9) \times 10^{-18}$	5.4×10^{-18d}	2.73×10^7	2.86
320	0.804	17	$(1.6_4 \pm 1.1_4) \times 10^{-17}$	1.4×10^{-17d}	1.21×10^7	2.48
400	0.780	14	$(1.9_2 \pm 0.8_3) \times 10^{-16}$	2.5×10^{-16d}	2.93×10^6	1.40
500	0.759	11	$(2.2_9 \pm 0.6_7) \times 10^{-15}$	2.7×10^{-15d}	5.84×10^5	0.88
700	0.734	8	$(5.2_8 \pm 1.2_2) \times 10^{-14}$	4.3×10^{-14d}	4.65×10^4	
1000	0.715	6	$(6.5_3 \pm 0.8_7) \times 10^{-13}$	3.9×10^{-13d}	1.30×10^4	0.53

^a Units of $\text{cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$. ^b Results obtained indirectly from MCQT calculations on the same potential energy surfaces. ^c Results taken from ref 8 (Persky) and corrected for the appropriate value of $f(T)$ [see text]. ^d Results taken from ref 9 (Johnson and Winter).

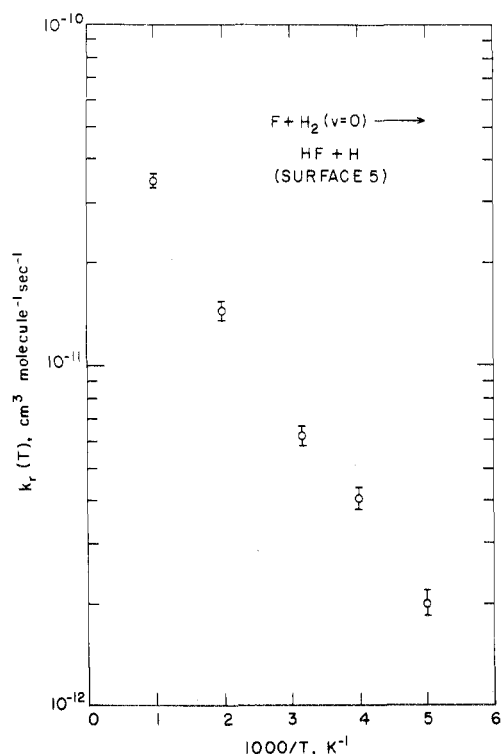


Figure 3. Calculated rate constant and estimated error for the reaction $F + H_2(v=0) \rightarrow HF + H$ at several temperatures. Importance sampling was used in the selection of the collision energy and the impact parameter. The potential energy surface employed was Muckerman's Surface 5.¹⁰

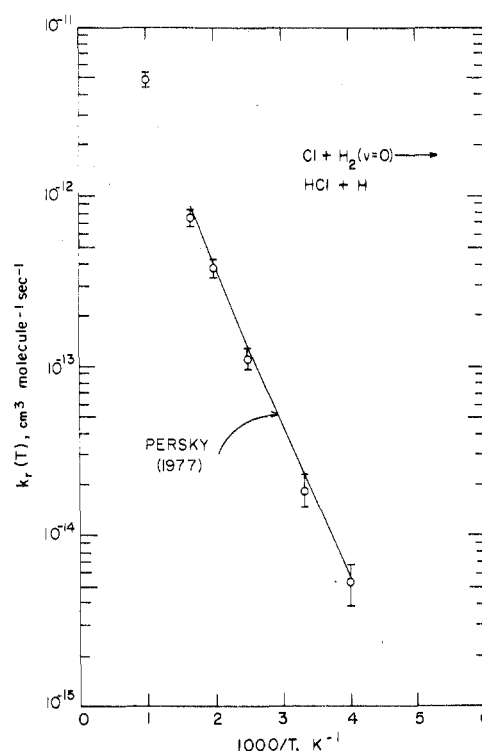


Figure 4. Calculated rate constant and estimated error for the reaction $Cl + H_2(v=0) \rightarrow HCl + H$ at several temperatures. Importance sampling was as in Figure 3. The solid line connects the (corrected) values reported by Persky⁸ using the same potential energy surface.

a limitation of the general form of the importance sampling function.

Having established diagnostics for the quality of importance sampling of the collision energy (as well as impact parameter⁵), we proceeded to calculate rate constants for each of the three systems under consideration at several temperatures. Each computed rate constant was based on a sample of 1000 trajectories with importance sampling of the initial impact parameter and collision energy. Crude Monte Carlo sampling was used in the selection of all other initial conditions. The collision energy was selected from the distribution $f_m(E_{\text{rel}}/kT)$ for some estimate of the

optimum value of m as discussed in section II.B. This was accomplished by generating a random value of u_{rel} uniformly distributed in the interval (0,1), and using the (iterative) Newton-Raphson method described in the Appendix to obtain the corresponding $E_{\text{rel}}(u_{\text{rel}})$. The impact parameter was selected by choosing a random value of u_b and computing $b(u_b)$ from the relation⁵

$$b = b_{\text{max}} \left\{ \cos \left[\frac{1}{3} \cos^{-1} (1 - 2u_b) + \frac{4\pi}{3} \right] + \frac{1}{2} \right\} \quad (28a)$$

which corresponds to the importance sampling function

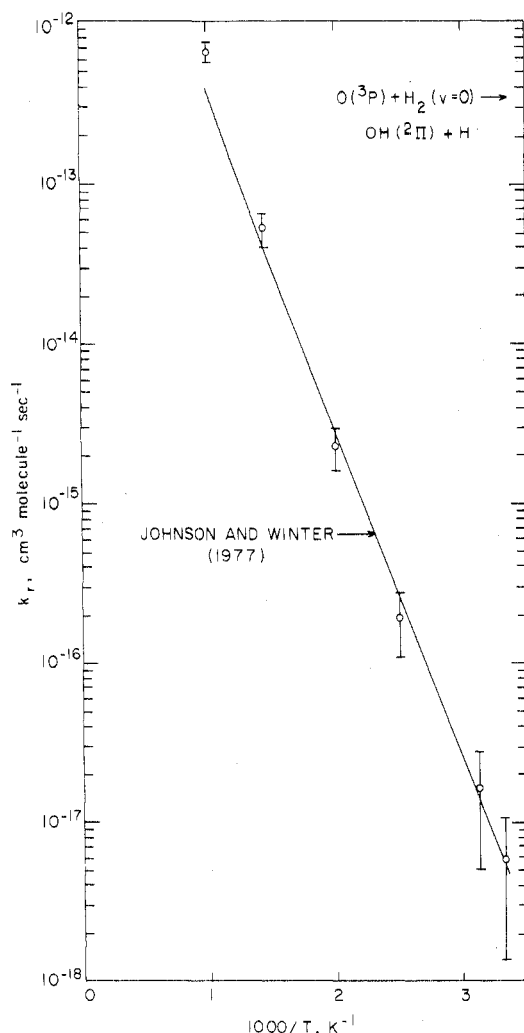


Figure 5. Calculated rate constant and estimated error for the reaction $\text{O}(^3\text{P}) + \text{H}_2(v=0) \rightarrow \text{OH} + \text{H}$ at several temperatures. Importance sampling was as in Figure 3. The solid line connects the values reported by Johnson and Winter⁹ using the same potential energy surface.

TABLE II: Comparison of Several Quantities from Two Calculations of the Rate Constant for the Reaction $\text{F} + \text{H}_2(v=0) \rightarrow \text{HF} + \text{H}$ at $T = 300 \text{ K}$

quantity	importance sampling, $m = 3$	standard (crude Monte Carlo) sampling
N	1000	3441
N_r	264	224
$\sum_{j=1}^{N_r} w_j$	57.1	224
$\langle P_r \rangle$	0.0571	0.0651
$\Delta \langle P_r \rangle$	0.0040	0.0042
$\Delta \langle P_r \rangle / \langle P_r \rangle$	0.0693	0.0645
DS_n, eu	0.38	1.84
$\tilde{N}$	3441	(2990) ^a

^a This value is not computed using eq 27, but is the estimated number of trajectories based on the value of $\langle P_r \rangle$ from the crude Monte Carlo calculation which would be required to obtain the relative error of the importance sampling calculation. Here $\tilde{N}_c = (1 - \langle P_r \rangle) / (\langle P_r \rangle R^2)$, where R is the relative error from the importance sampling calculation.

$$P_r^o(b) = \begin{cases} 3(1 - b/b_{\max}) & b \leq b_{\max} \\ 0 & b > b_{\max} \end{cases} \quad (28b)$$

The values 2.0, 1.6, and 1.4 Å were used for $b_{\max}$ in the $\text{F} + \text{H}_2$, $\text{Cl} + \text{H}_2$, and $\text{O}(^3\text{P}) + \text{H}_2$ calculations, respectively.

The results from these calculations are presented in

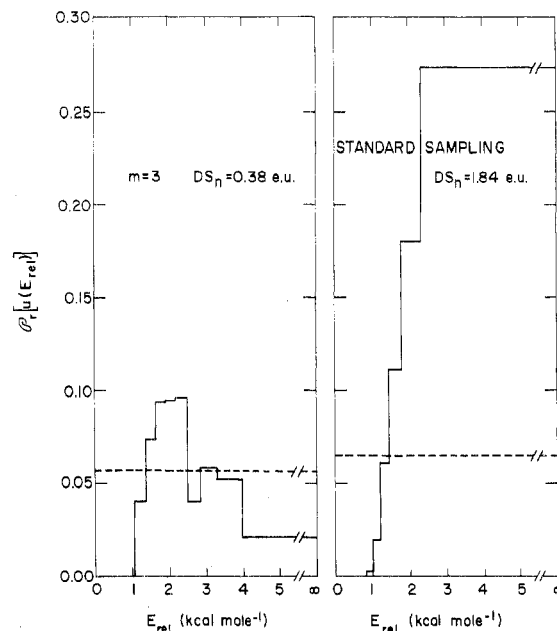


Figure 6. Plot of the histogrammic representations of $P_r[u(E_{\text{rel}})]$ (solid lines) for ten equal increments of u from an importance sampling calculation (left-hand panel) and a standard (crude Monte Carlo) sampling calculation (right-hand panel) of the rate constant for the $\text{F} + \text{H}_2(v=0)$ system at $T = 300 \text{ K}$. The dashed lines indicate the values of $\langle P_r \rangle$ obtained from the two calculations. The DS_n values are also indicated.

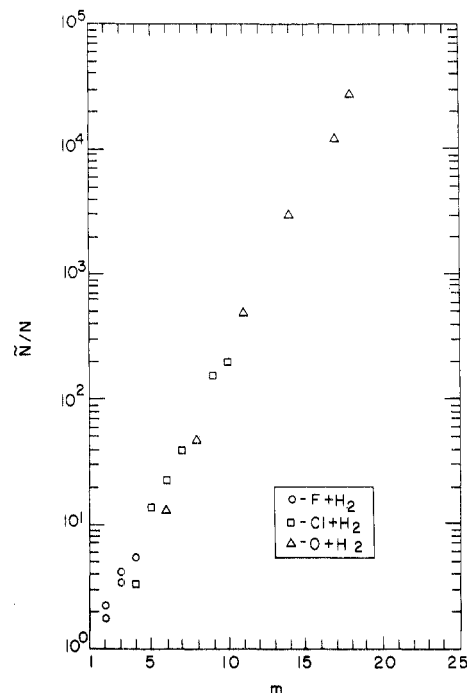


Figure 7. A plot of all estimated values of $\tilde{N}/N$ taken from Table I vs. the optimum value of m . Data for all systems at all temperatures are combined. The striking linear correlation between $\log(\tilde{N}/N)$ and m is independent of both system and temperature.

Table I. Also listed in the table are values of rate constants (calculated indirectly) taken from the previous studies of the $\text{Cl} + \text{H}_2$ ⁸ and $\text{O}(^3\text{P}) + \text{H}_2$ ⁹ systems, DS_n , $\tilde{N}$, and the multiple surface coefficient, $f(T)$.¹¹ Persky's reported rate constants⁸ have been corrected to account for the deviation of $f(T)$ from the value of 0.5 which he used. The DS_n values are seen to be larger at lower temperatures and for the larger values of m as was discussed above. These rate constant data are also shown in Figures 3–5 for the $\text{F} + \text{H}_2$, $\text{Cl} + \text{H}_2$, and $\text{O}(^3\text{P}) + \text{H}_2$ systems, respectively. The points with error bars represent the present (im-

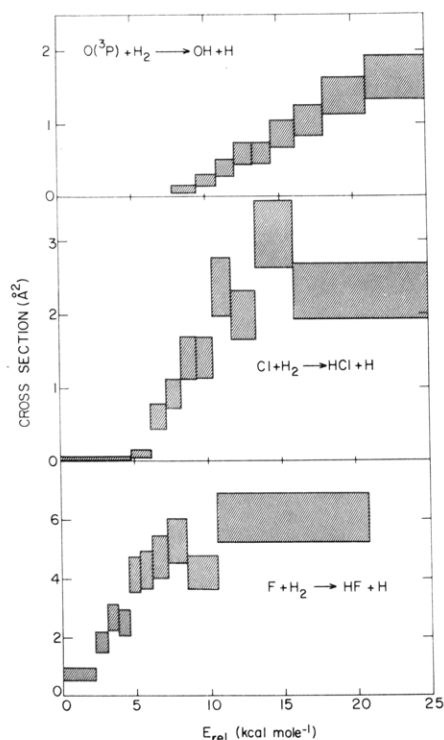


Figure 8. Plot of histogrammic representations of $\sigma_r[u(E_{rel})]$ for ten equal increments of u for the $F + H_2(v=0)$ [lower panel], $Cl + H_2(v=0)$ [middle panel], and $O(^3P) + H_2(v=0)$ [upper panel] systems obtained from importance sampling calculations of rate constants for these reactions at $T = 1000$ K. The excitation functions plotted correspond to a rotational temperature of the $H_2(v=0)$ molecule of 1000 K.

portance sampling) results and the lines (in Figures 4 and 5) are drawn through the (indirectly determined) rate constants reported in the previous studies of these systems. Agreement with the previous calculations is generally excellent, and the present results were obtained with at least one order of magnitude fewer trajectories. Note that average reaction probabilities as small as 10^{-7} are estimated with reasonable accuracy from a sample of 10^3 trajectories when importance sampling is employed.

The ratio $\tilde{N}/N$, with $\tilde{N}$ defined by eq 27 and $N = 1000$, may be used to estimate the efficiency of the importance sampling approach to the calculation of rate constants compared to similar calculations employing crude Monte Carlo sampling. That $\tilde{N}$ is at least a rough estimator for the number of samples required using crude Monte Carlo sampling to obtain the same relative error as that from an importance sampling calculation is demonstrated for a case in which $\tilde{N}$ is not prohibitively large [the $F + H_2(v=0)$ system at $T = 300$ K] in Table II. Listed in the table are several quantities associated with the importance sampling calculation based on a sample of 1000 trajectories and their corresponding values from a crude Monte Carlo calculation with a sample of 3441 trajectories, the value of $\tilde{N}$. Although the agreement between the two calculations is not particularly good, the two sets of results are not inconsistent; the value of $\tilde{N}$ is within 15% (3441 vs. 2990) of the value of $\tilde{N}_c$ (defined in Table II) based on the crude Monte Carlo results. Figure 6 presents histogrammic plots of $\mathcal{P}_r[u(E_{rel})]$ with equal increments of u for these two calculations which show the smaller variance of the importance sampling integrand.

The values of $\tilde{N}$ from Table I are used in a plot of $\tilde{N}/N$ as a function of the optimum value of m in Figure 7. Data from all systems at all temperatures are plotted in the same figure. A striking linear correlation is observed between

$\log(\tilde{N}/N)$ and m which indicates an exponential increase in the relative efficiency of the present importance sampling approach as the optimum value of m increases. This property appears to be independent of the system and temperature.

Finally, we turn to extracting information about the reaction cross section as a function of collision energy from the direct importance sampling calculation of rate constants. Histogrammic representations of this function for each of the three systems studied at each temperature (the cross section obtained depends on the internal temperature of the diatomic molecule) can be obtained using eq 24 and 25. Figure 8 shows such results for each of the three systems at $T = 1000$ K. Although these histogrammic excitation functions exhibit rather low resolution, they are entirely consistent with previously computed cross sections using the same potential energy surfaces.⁸⁻¹⁰ In fact, plots such as those in Figure 8 obtained from the direct calculation of a rate constant provide a very efficient method for obtaining an overview of an entire excitation function.

IV. Concluding Remarks

In the preceding sections we have discussed the general features of importance sampling as a variance reduction technique in the Monte Carlo evaluation of (multidimensional) integrals, and have applied this technique to the direct MCQT calculation of bimolecular gas phase rate constants. Numerical examples were provided to demonstrate the accuracy and greatly increased efficiency of such an approach over both an indirect MCQT method, which first requires the calculation of excitation functions, and a direct MCQT calculation using crude Monte Carlo sampling. In the examples cited, a general importance sampling function for the collision energy having a single (integer) parameter was employed. Guidelines for specifying a good choice of this parameter were offered, and diagnostics for determining after a calculation has been carried out whether the choice of the sampling parameter were optimum and whether the form of the energy importance sampling function were appropriate. It was also shown that the calculation of a rate constant using importance sampling on the selection of the collision energy can provide an overview of the excitation function for the reaction.

The potentialities of the importance sampling technique for MCQT calculations are not limited to those applications made in the present work. A natural extension of the approach taken in the determination of the excitation function from a direct rate constant calculation is the determination of such other properties as the opacity function, product angular distributions, and product internal state distributions. Expressions for these quantities may be derived straightforwardly following the procedure of Faist et al.⁵ The Tolman activation energy for the reaction can also be determined by a rate constant calculation at a single temperature.^{12,13}

Hopefully the results obtained in the present work will encourage further research in the application of the importance sampling technique to MCQT calculations. Clearly the importance sampling function for the collision energy used here is not optimum for all systems at all temperatures, and other choices for this function may result in even greater efficiency. In calculations involving complicated potential energy functions (e.g., evaluating the potential energy by diagonalizing a large DIM Hamiltonian matrix¹⁴), the increased efficiency offered by the importance sampling technique can provide a far more reliable result for a given number of computed trajectories.

TABLE III: Probabilities for Sampling Collision Energy Less Than Certain Values for Various Values of m Using the Importance Sampling Function Defined by Eq 17

m	$(m - m^{1/2})kT$	mkT	$(m + m^{1/2})kT$
1	0	0.264	0.594
2	0.022	0.323	0.663
3	0.040	0.353	0.695
4	0.053	0.371	0.715
5	0.062	0.384	0.728
6	0.069	0.394	0.738
7	0.075	0.401	0.746
8	0.080	0.407	0.752
9	0.084	0.413	0.758
10	0.087	0.417	0.762
11	0.090	0.421	0.766
12	0.093	0.424	0.769
13	0.095	0.427	0.772
14	0.098	0.430	0.775
15	0.100	0.432	0.777
16	0.101	0.434	0.779
17	0.103	0.436	0.781
18	0.104	0.438	0.783
19	0.106	0.439	0.784
20	0.107	0.441	0.786
21	0.108	0.442	0.787
22	0.109	0.444	0.788
23	0.110	0.445	0.789
24	0.111	0.446	0.791
25	0.112	0.447	0.792

Appendix. Initialization of the Collision Energy

The uniform variable, u , the value of which is selected randomly, is related to the reduced collision energy by eq 18f for the choice of importance sampling function given by eq 17 and used in the present work. As the function $u(x)$ cannot be inverted to obtain a closed form expression for $x(u)$, the appropriate value $x^* = x(u^*)$ for some specific value u^* of u must be determined numerically. This may be accomplished by determining the root of the equation

$$\mathcal{G}(x) \equiv u(x) - u^* = 0 \quad (\text{A.1})$$

Now the derivative of $\mathcal{G}(x)$ is simply the function $f_m(x)$ as defined in eq 18d. Therefore, upon applying the Newton-Raphson iterative method to obtain the root of eq A.1, the successive approximations to x^* are

$$x_{i+1} = x_i - \frac{\mathcal{G}(x_i)}{\mathcal{G}'(x_i)} \quad (\text{A.2})$$

$$= x_i - \frac{m!}{x_i^m} e^{u^*}(1 - u^*) + \frac{m!}{x_i^m} \sum_{l=0}^m \frac{x_i^l}{l!} \quad (\text{A.3})$$

The iteration is always convergent, approaching the correct value of x^* from one side, if the initial guess is taken to be the inflection point in $u(x)$, i.e.

$$x_0 = m \quad (\text{A.4})$$

An entirely different approach to selecting appropriate values of x is to use the von Neumann rejection technique. Here it is necessary to transform the variable x in the interval $(0, \infty)$ to some new variable y in the interval $(0, 1)$. This is accomplished by defining

$$y \equiv 1 - \frac{1}{x + 1} \quad (\text{A.5})$$

The Jacobian of transformation, dx/dy , must then be included as a multiplicative factor in the expression for the transformed distribution function $g_m(y)$, i.e.

$$g_m(y) = f_m(x(y)) \frac{dx}{dy} \quad (\text{A.6})$$

$$g_m(y) = f_m(x(y))/(1 - y)^2 \quad (\text{A.7})$$

Appropriate values y^* of y may be obtained by rejection using the distribution function $g_m(y)$. The corresponding value of x^* is readily obtained by solving eq A.5 for x . This rejection procedure is not as efficient, especially for large values of m , as the Newton-Raphson iterative method discussed above because $g_m(y)$ becomes very sharply peaked and many trial values of y^* are therefore rejected before an acceptable one is found. For other choices of importance sampling function, however, the rejection technique may be the only practical alternative.

References and Notes

- (1) (a) Research performed at Brookhaven National Laboratory and supported by the U.S. Department of Energy, Office of Basic Energy Sciences. (b) Aerodyne Research, Inc., Bedford Research Park, Bedford, Mass. 01730.
- (2) See, for example, J. T. Muckerman, *J. Chem. Phys.*, **54**, 1155 (1971), and ref 8 and 9.
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- (11) (a) D. G. Truhlar, *J. Chem. Phys.*, **56**, 3189 (1972); (b) J. T. Muckerman and M. D. Newton, *ibid.*, **56**, 3191 (1972).
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- (14) See, for example (a) J. C. Tully and R. K. Preston, *J. Chem. Phys.*, **55**, 562 (1971); (b) P. A. Whitlock and J. T. Muckerman, manuscript in preparation; (c) P. A. Whitlock, Ph.D. Thesis, Wayne State University, Detroit, Mich., 1976.

Discussion

C. SLOANE (General Motors Research Laboratories). Since the rate coefficient, $k(T)$, depends on an integral over the overlap of the Boltzmann energy factor and the reaction cross section, your importance sampling affects the accuracy of $k(T)$ by most precisely determining the cross section in the energy region where the overlap is significant. Often this is the vicinity of the threshold. Thus, in your example, 1000 energetically importance sampled trajectories gave a $k(T)$ equivalent to that found with 80 000 conventionally sampled trajectories. It might be noted, however, that much previous work has been done at energies higher than threshold and extrapolated downward because of the often greater cost of calculations near threshold energies. Thus, one may save over an order of magnitude on the number of trajectories needed to evaluate $k(T)$ by using importance sampling *without* a corresponding savings in cost (computer time) because of the potentially greater cost of trajectories at lower energies.

J. T. MUCKERMAN. There are two factors which are important in this regard. The first is the accuracy of any such extrapolation. How does one ensure extrapolating to an accurate threshold? In the importance sampling procedure we have described, one need not know the exact threshold; a reasonable estimate will suffice. The resulting rate constant does not depend on any extrapolated threshold and should consequently be more accurate.

Secondly, it should be noted in our procedure, the inner inflection point of the importance sampling function, defined by eq 17, is placed slightly below the estimated threshold energy of the reaction. As mentioned previously, this inner inflection point

occurs at the energy $(m - m^{1/2})kT$. The most probable collision energy is mkT , and the outer inflection point occurs at $(m + m^{1/2})kT$. Table III indicates the probability of selecting a collision energy below each of these three characteristic energies for various values of m . In our calculation of the rate constant for the $O + H_2$ reaction at 320 K, for example, we sampled collision energies lower than 8.2 kcal mol⁻¹ only 10% of the time, collision energies between 8.2 and 13.4 kcal mol⁻¹ 68% of the time (energies near 10.8 kcal mol⁻¹ were more probable), and collision energies above 13.4 kcal mol⁻¹ the remaining 22% of the time. The energy threshold for the reaction is near 8 kcal mol⁻¹.

In summary, we feel that our procedure avoids introducing any error in the computed rate constant arising from an assumed or extrapolated threshold, yet does not sample excessively in the region of the threshold.

DONALD G. TRUHLAR (University of Minnesota). The paper by Muckerman and Faist shows clearly that importance sampling is useful for calculating thermal rate constants by the Monte Carlo trajectory method. Blais and I have found that stratified sampling is also useful for this purpose. (N. C. Balis and D. G. Truhlar, *J. Chem. Phys.*, to be published.)

I. W. M. SMITH (University of Cambridge). Dr. J. C. Naylor and I are engaged in calculations on atom-transfer reactions which are designed to overcome two major defects of the conventional quasi-classical trajectory method. The first of these, referred to in Muckerman's presentation, is that frequently only a small fraction of trajectories lead to reaction, even above threshold, so the calculations are inefficient and expensive. Secondly, internal energies are usually chosen to correspond to quantum levels in the molecular reagents, but thereafter the classical equations of motion are solved and the energies are no longer "restricted".

In our calculations, trajectories are started at the critical dividing surface defining the transition state. Energies are selected which correspond to quantum levels at these configurations and trajectories are computed forward and backward in time to identify those which pass through the transition state more than once. The work has two main objectives: (1) to discover for which purposes our classical trajectory plus transition state method provides answers at least as good as those from conventional quasi-classical trajectories, at reduced cost; (2) to explore the connection by classical trajectories of quantized levels in the transition state and those in the phase space of the reagents.

Halogen Recombination-Dissociation Reactions. Current Status[†]

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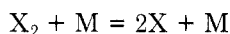
Chemistry Department, University of Toronto, Toronto, Ontario, Canada (Received July 24, 1978)

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Recombination and dissociation rate constants for halogens, obtained by a variety of experimental techniques, are assembled and compared. In general, more accurate and precise rate constants are obtained by more recent investigators, who learned from the experiences of their predecessors. Surprisingly, a large amount of data on recombination of halogen atoms, in the presence of even the simplest third bodies, is still not available. On the other hand, much of the available data can now be considered quite reliable. Trajectory calculations yield recombination rate constants in fair agreement with experiments.

Introduction

Recombination-dissociation reactions of homopolar diatomic halogen molecules have been studied extensively over the past 40–50 years, and an appreciable amount of experimental and theoretical data is now available. The present manuscript deals with these reactions studied in the presence of a large excess of simple third bodies, such as inert gases, M:



The purpose of this review is to (1) assemble experimental data obtained using various techniques and compare them with each other, (2) determine, whenever possible, major causes of experimental errors, (3) present, whenever possible, summaries of the most reliable experimental data over a wide temperature range, (4) determine which recombination-dissociation reactions need further study, either because of lack of experimental data or inaccuracy of available experimental results, and (5) compare experimental rate constant data to classical trajectory results.

Bromine

$Br_2 + M \rightleftharpoons 2Br + M$ reactions have been studied most extensively, especially for $M = Ar$. The earliest work was

performed using the stationary photolysis technique.^{1–3} The results obtained by this technique over 40 years ago¹ have proved remarkably accurate. To our knowledge, there has been no further work using this technique. Early flash photolysis data⁴ showed some persistent systematic error. The results⁴ can be separated into two groups. During the first year of their work, the authors obtained relatively low values for flash photolysis rate constants, which agreed well with stationary photolysis results. Thereafter, there was a sudden increase in the rate constants which persisted for about 1 year, during which time many measurements were made. However, at the end, these authors were able to obtain again their "low" value rate constants.

Subsequent flash photolysis studies^{5–13} of these reactions gradually improved in precision, and tend to agree with each other and with "low" data.⁴ For this reason, we are inclined to believe that the "high" recombination rates obtained in early work⁴ are probably in error. Our experience indicates that most impurities introduced into the system tend to increase the apparent rate constant in experiments such as those described in ref 4. This is because inert gases and simple diatomic third bodies are less efficient than any of the more complex polyatomic impurities. In addition, in flash photolysis, various impurities and their fragments may initiate chain reactions, which would yield spuriously high recombination rates. Currently, at least at room temperature, flash photolysis results agree with each other within 10–20%, which is a typical standard deviation quoted by various authors.

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